

NOTEPAD APPLICATION



2023-2024
A PROJECT REPORT

Submitted by

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In partial fulfilment for the award of the degree of

BACHELOR OF COMPUTER APPLICATIONS

Under the guidance of

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Department of Computer Science

Certificate

Certified that the project work entitled

NOTEPAD APPLICATION

.....is a bonafied work carried out by

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in partial fulfilment for the award of degree of Bachelor of Computer Applications of the Mangalore University during the year 2023-24. The project report has been approved as it satisfies the academic requirements of project work prescribed for the Bachelor of Computer Applications.

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DECLARATION

We hereby declare that this project work entitled “**NOTEPAD APPLICATION**” has been prepared by us during the year 2023-2024 under the guidance of **Mrs. VIJAYALAKSHMI SHETTY**, Department of Computer Science Bhandarkar’s Arts and Science College, Kundapura in the partial fulfilment of BCA degree prescribed by the Mangalore University.

We also declare that this project is the outcome of our own effort, that it has not been submitted to any other university for the award of any degree.

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Our sincere thanks for all faculty members of Computer Science Department. We are thankful to our parents for their encouragement towards the project. Last but not the least, we whole heartly appreciates the co-operation of our friends.

Thank You,

Project Team:

- ABHIJNA
- RASHMITHA
- SINCHANA

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1.INTRODUCTION OF THE PROJECT

1.1Abstract:

Notepad is a basic text editor for Microsoft Windows. It has been included in all versions of Microsoft Windows since Windows 1.0 in 1985. It can be used to create simple documents. The most common use for Notepad is to view or edit text (.txt) files. Many users find Notepad a simple tool for creating web pages because Notepad supports only very basic formatting, you cannot accidentally save special formatting in documents that need to remain pure text. Notepad includes modules like File, Edit, Format, Help.

The File menu gives the options like new, open, save, save as, exit. The Edit menu allows us to copy, cut, paste, undo and redo text. The Format menu allows us to view status bar and the Help menu gives the information about Notepad.

1.2.Background and Introduction of sponsor/client and problem statements:

A notepad is a simple tool that lets you write down and organize your thoughts, ideas, or important information. It's like having a digital version of a paper notebook where you can jot down anything you need to remember.

The background of a notepad dates back to the time when people used to carry around small notebooks or pads of paper to write down notes, reminders, or to-do lists. These physical notepads served as handy companions for keeping track of important information.

With the advent of computers and digital technology, notepads evolved into software applications that run on computers, tablets, and smartphones. Digital notepads offer many advantages over their physical counterparts, such as the ability to easily edit and rearrange notes, search for specific information, and access notes from anywhere with an internet connection.

1.3. Literature review:

A notepad is a simple tool that lets you write down and organize your thoughts, ideas, or important information. It's like having a digital version of a paper notebook where you can jot down anything you need to remember.

For students involved in the project, it provides an opportunity to apply theoretical knowledge into practical implementation. They can gain hands-on experience in software development, including programming, user interface design, and testing. Some notepad projects allow for collaboration, where multiple users can work on the same document simultaneously. This fosters teamwork and makes it easier to share information or work on group projects.

Overall, a notepad project simplifies the process of note-taking, improves organization and productivity, facilitates learning, promotes collaboration, and offers digital accessibility, making it a valuable tool for various personal, academic, or professional purposes.

1.4.Aims and Objectives:

The aims and objective of a Notepad project are:

- User-Friendly Interface.
- Note Organisation.
- Cross-Platform Compatibility.
- Syncing Capability.
- Security.
- Customization Options.
- Collaboration Features.
- Search Functionality.
- Performance Optimization.
- Feedback And Improvement.

1.5. Scope of the Project:

The Notepad project's scope encompasses personal note-taking, educational use, and professional applications. It serves as a versatile tool for individuals, students, and professionals alike, facilitating tasks such as jotting down ideas, organizing information, and drafting documents efficiently. The project's flexibility allows for various customization options to suit different user needs and preferences.

1.6. Proposed Methodology:

- **File:** In a file module, it handles interactions with files. Users can Open, Read, Write, Create and Close a file.
- **Edit:** Edit module handles operations related to manipulating text within the notepad interface. Like, Text selection, Cut, Copy, Paste, Undo, redo, Time and Date, Find and Select All.
- **Help:** The help module serves as a resource for users to access information, instructions, and assistance related to application's functionality and usage.
- **Font:** It allows users to customize the appearance of the text they are editing like, Font size, type, style (bold, italic).
- **View:** The view module include Status-Bar and Word-wrap.

1.7. Proposed Solution and Anticipated Result:

i) Text Editing Features:

- Ability to create new text documents.
- Ability to open existing text files.
- Ability to save text documents.
- Ability to undo and redo actions.

ii) Text Manipulation:

- Copy, cut, and paste functionality.
- Find and replace text within documents.

iii) File Management:

- Ability to organize documents into folders.
- Support for renaming, deleting and moving files.
- Option to export documents in different formats (ex: plain text, HTML).

iv) User Interface:

- Intuitive and user-friendly interface.
- Customizable themes and appearance options.
- Responsive design for different screen sizes.

v) Keyboard shortcut:

- Support for common keyboard shortcuts for efficiency in text editing (ex: Ctrl+C for copy, Ctrl+V for paste, Ctrl+N for new and so on).

vi) Find:

- Ability to search for specific text within a document.
- Filter functionality to quickly find documents based on criteria such as date modified or file type.

vii) Accessibility:

- Support for accessibility features such as screen readers and keyboard navigations.
- Adjustable font sizes and colors for better readability.

viii) Customization:

- Setting menu to allow users to customize preferences such as default font, default save locations etc.

1.8. Schedule of activities:

Activity	Tentative Date
1.Introduction of the project and Objectives of the project	Week1
2. Proposal submission	Week2
3. Completion of File Menu	Week3
4. Completion of the Edit Menu	Week4
5. Completion of the Help Menu	Week5
6. Completion of the Format Menu	Week6
7. Completion of the shortcut keys	Week7
8. Testing	Week8
9. Documentation	Week9
10. Finalizing the project	Week10

1.9.Requirements:

1.9.1. Hardware Interface:

The physical computer resources (hardware) needed for your Notepad project include:

i)Processor: An Intel Pentium or above. RAM with 4GB or above.

ii) Hard Disk: At least 320GB of storage.

iii)Display Type: A standard PC display.

iv) Keyboard: Use a PC/AT Enhanced PS/2 keyboard (110/10-key).

v) Mouse: Opt for a First/Pilot Mouse Serial (c48).

vi)Input Device: Both mouse and keyboard are essential.

1.9.2. Software Interface:

i)Operating System: For better performance, use **Windows 8 or above**.

ii) Front End and Back End: For the front end and back end usage of Python.

iii) Text Editor: For the purpose of editing of text usage of VS code.

1.9.3. Communication Interface:

Not Applicable.

1.10.About project

Notepad is a basic text editor that's built into Windows. It is excellent for writing relatively short text documents that you want to save as plain text.

In the world of computers, among all the fancy software out there, there's a simple yet super important tool: Notepad. Since it first came out, Notepad has been a common sight on Windows computers, quietly helping people with writing and editing text every day. Even though it looks

plain, Notepad is loved by people everywhere because it's easy to use and gets the job done.

This project is all about understanding what makes Notepad special and why it's still so useful today. We'll take a close look at what it can do, where it came from, and how people use it in their lives. From its beginnings to how it fits into today's tech world, Notepad is a key tool for professionals, students, and anyone who needs to work with text.

By looking into Notepad, we'll also learn about the importance of keeping things simple when designing software. Notepad's been around for a long time, but it still proves that simplicity is a big deal in the digital world. Come along as we explore the timeless charm and practical power of Notepad, and discover why simplicity is still king in today's tech age.

1.11.1 Modules that involved in NOTEPAD:

1. File Module:

The *Edit* menu offers a few choices, but again, everything on this menu should be familiar to anyone who has used Windows. All the *Edit* choices have associated keyboard shortcuts. Note that most of the commands are greyed out until there is text selected in the *Notepad* window.

The first item on the *Edit* menu is *cut,copy,paste* which can be useful when you are editing the document. What appears in this place depends on what you have been doing. This is the core functionality of Notepad. It allows users to input, edit, and manipulate text. Users can type, delete, copy, cut, and paste text within the editor.

2. Edit Module:

Notepad includes features for managing text files

The choices you have in the *File* menu are *New,, Open, Save, Save As, Exit*. As you can know, many of these commands have keyboard shortcuts as well.

Creating and saving text documents in *Notepad* is simple: open *Notepad*, start typing, and then edit and format the text as you see fit. Once you are finished, use the *Save As* command to save your work. The default folder is the [OneDrive](#) folder in Windows 10, and the *My Documents* folder in Windows 7. You can change this quite easily: use the *Save As* command, browse to your preferred folder, and click *Open*. *Notepad* will remember your choice. Keep in mind that your files are saved with a *.txt* extension and in plain text.

3. Font Module:

The *Font* choice is self-explanatory: it offers you a list of all your installed fonts and the option to use bold, italic, and the like. However, unlike the way it works in programs like *Microsoft Word*, a change of font immediately affects the entire document. You cannot use one font in one part of the document and another font somewhere else. It is all or nothing.

In the *Font* menu, there is a less familiar option available, the drop-down menu labelled *Script*. This lets you choose characters that are not available in the standard "Western" style fonts. The choices are *Western*, *Greek*, *Turkish*, *Times New Roman*, *Baltic* (not available in Windows 7), *Central European*, *Cyrillic*, and *Vietnamese* (not available in Windows 7).

4. View Module:

Find is used to search the text in the document, To search for a piece of text, select the *Find* option (keyboard shortcut *Ctrl + F*) depending on what you want. That makes *Notepad* open a box where you can type the text you're looking for in the "*Find....*" Field. It enhances productivity by facilitating text manipulation tasks.

5.Help Module:

- The help module serves as a resource for users to access information, instructions, and assistance related to application's functionality and usage.

2. SYSTEM DESIGN

System design is the process of defining the architecture, module interfaces and data for a system to satisfy specified requirements.

- The purpose of the design phase is to plan the solution of the problem specified by the requirement documents.
- This is the first step that moving from problem domain to the solution domain.
- The design of the system is essentially a blueprint or a plan for solution for the system.

2.1Functional decomposition

Functional decomposition is the process of taking a complex process and breaking it down into its smaller, simpler parts. Using functional decomposition large or complex functionalities are more easily understood. It is mainly used during project analysis phase, so each phase can be viewed as software. So, this has modular with some sub modules.

System software architecture

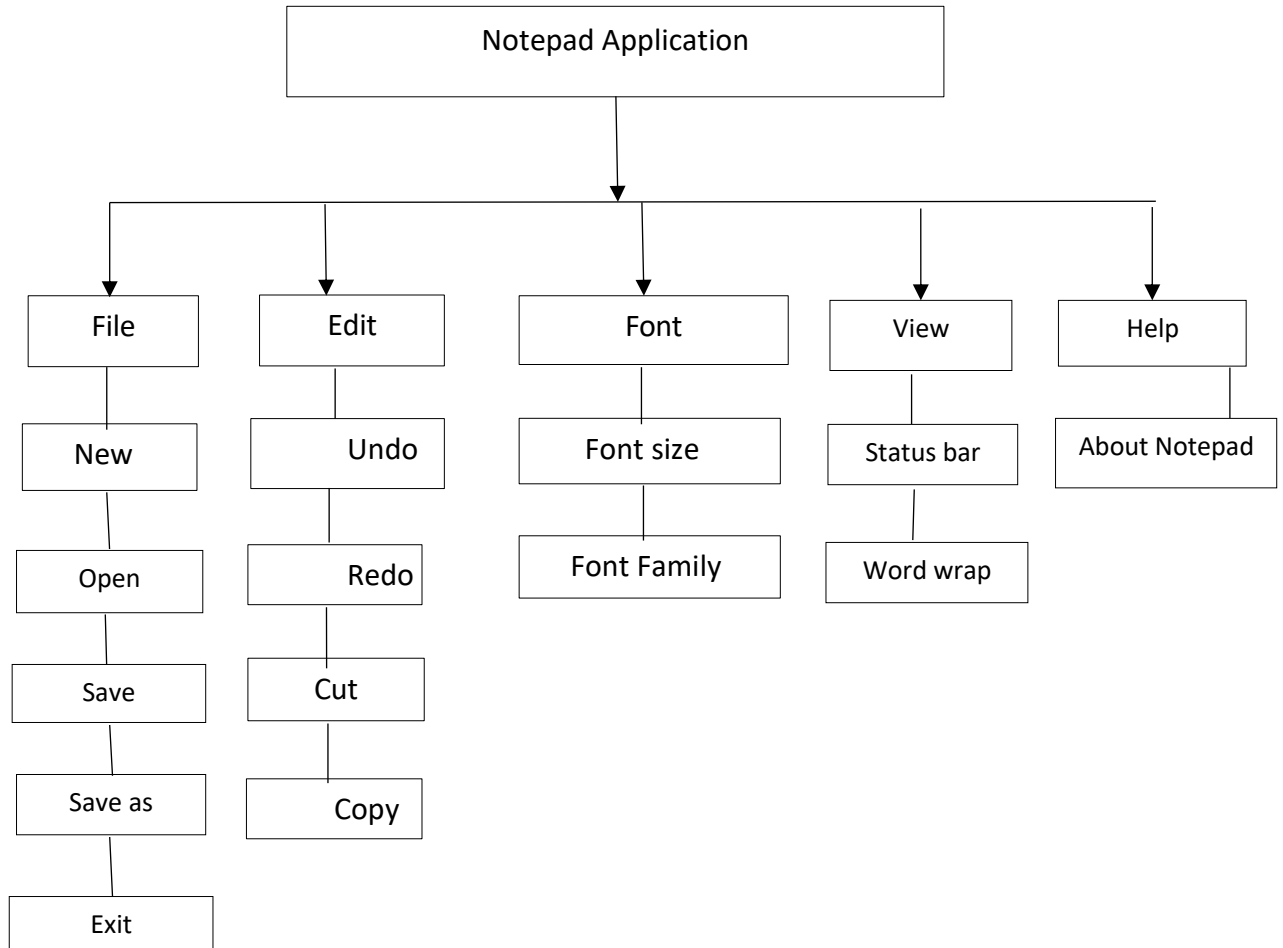


Fig 2.1 System software architecture

2.2. Description of programs

2.2.1. Context Flow Diagram (CFD)

In CFD entire system is considered as a single process. Content Flow Diagram shows all the external entities that interact with system and how the data flows between these external entities and system. External entity represented as rectangle. Data flows are represented by means of directed arrows.

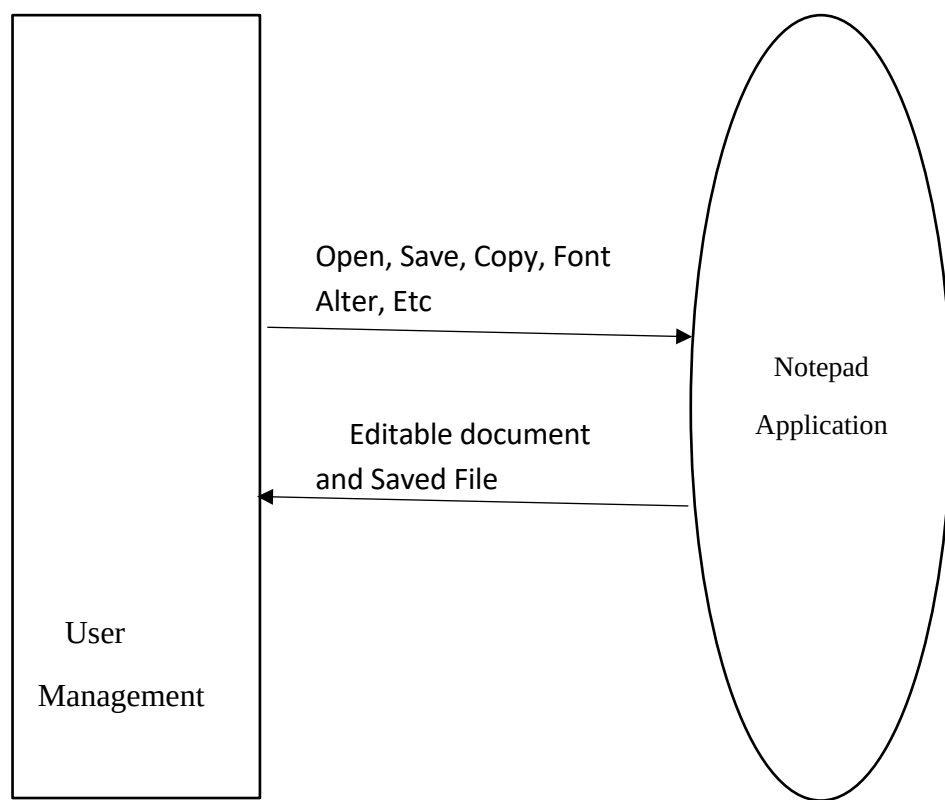


Fig2.2 Context flow diagram

2.2.2.Data Flow Diagram (Data Flow Diagram (DFD))

Data flow diagram shows the flow of data through system. Data flow diagram also called the data flow graphs. It views a system as a function that transforms the input into desired outputs. It aims to capture the transformation that taken place within a system to the input data so that eventually the output data is produced.

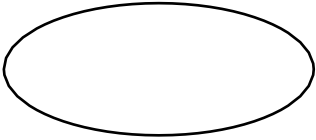
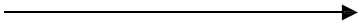
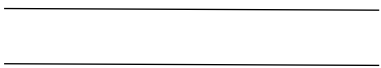
Symbols	Name	Description
	Process	It preforms transformation of data from one state to another.
	Source/Sink	It represents the external entity that may be either source or sink.
	Flow of data	It represents the flow of data source to destination.
	Data Source/Data Storage	It is the place where data is stored.

Table 2.3 Data Flow Diagram (DFD)

2.3.DFD For New:

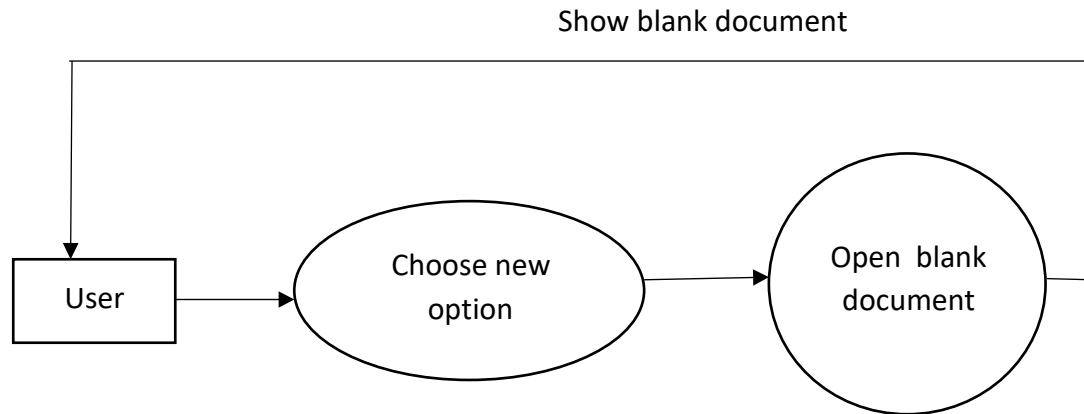


Fig 2.4 File

2.3.1 : NEW:

Input: Click for new option

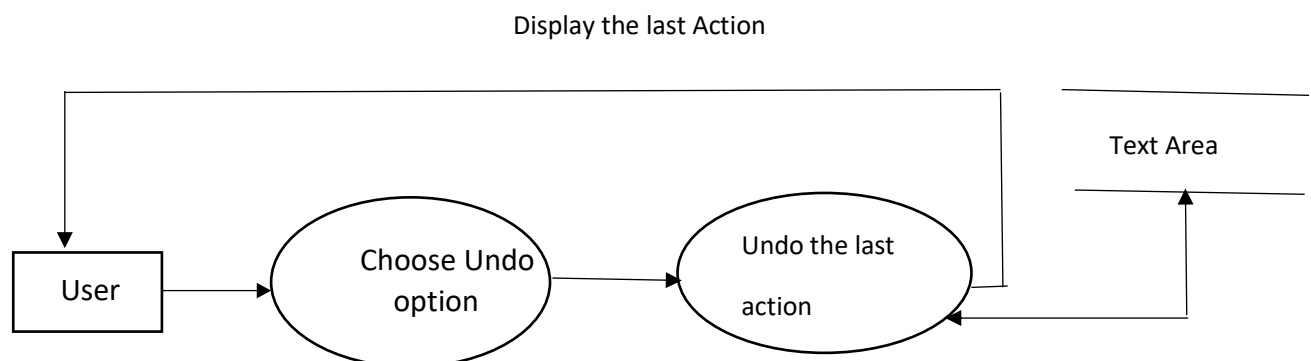
Process: it allow us to open blank page after saving an file if it exists

Output: Open a new black page.

Interface with other functional components: This is independent module.

Resource Allocation: Stored in the system.

2.4.DFD For UNDO :



2.5: UNDO:

Input: Click for an Undo option.

Process: In the event of error, it deletes the last stroke.

Output: It undo the action.

Interface with other functional components: This is independent module.

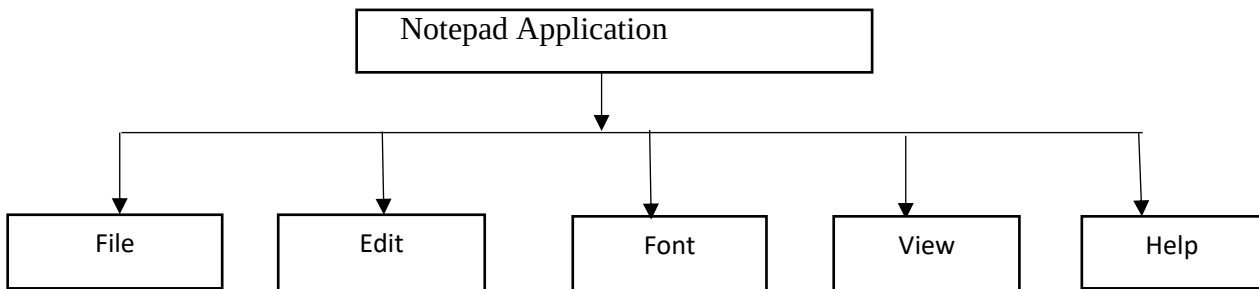
Resource Allocation: Stored in the system.

3.Detailed Design

3.1Introduction:

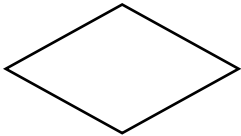
During detailed design, the internal logic of each module specified in system design is decided. During this phase further details of the module is decided. Design of each of the module is usually specified in a high-level description language which is independent of the language in which software eventually be implemented.

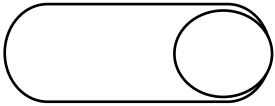
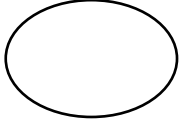
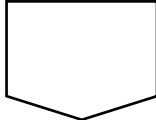
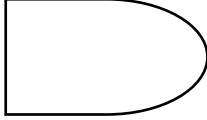
3.2Structure of the Software package:



3.3Flow chart:

Flow chart is graphical representation of solution to the given problems. A flowchart is pictorial representation of an algorithm, workflow or process. The diagrammatic representation illustrates a solution model to given problem. It uses the following symbol.

Symbol	Name	Purpose
	Terminator	It indicates the start and end process.
	Input/Output	Input/Output data.
	Decision	It represents comparison or question that determines an alternate path to be followed.
	Flow direction	Shows the direction of data flow.
	Processing	It represents manipulation, calculation or information process

	Direction access storage	File storage.
	Preparation (Looping)	An Instruction or Group of instruction.
	In-page	
	off-page	
	Delay	

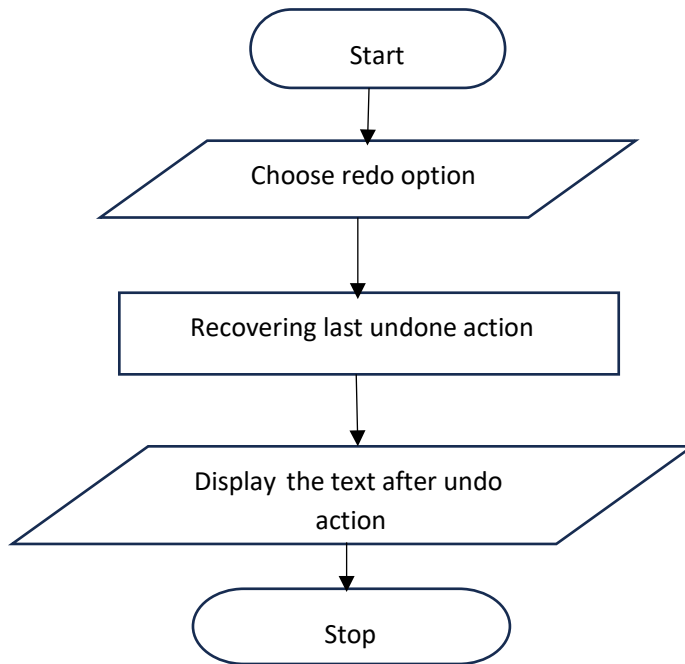
3.4Algorithm For REDO:

- Step1: Start
- Step2: Choose redo option.
- Step3: Recovering undone action.
- Step4: Display last undone action.
- Step5: Stop

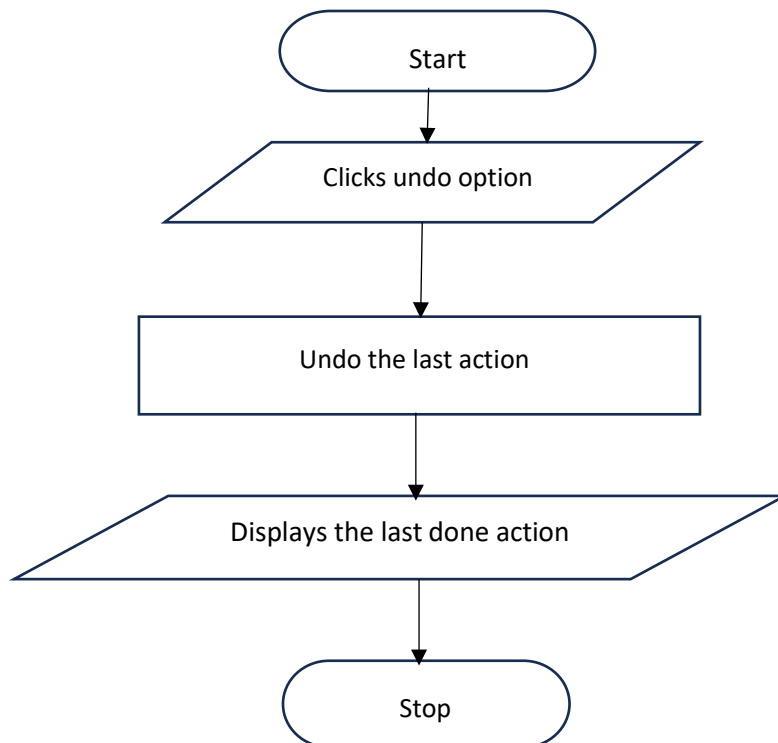
3.5Algorithm For UNDO:

- Step1: Start
- Step2: Choose undo option.
- Step3: Undo the last action.
- Step4: Display the text content to the current action.
- Step5: Stop

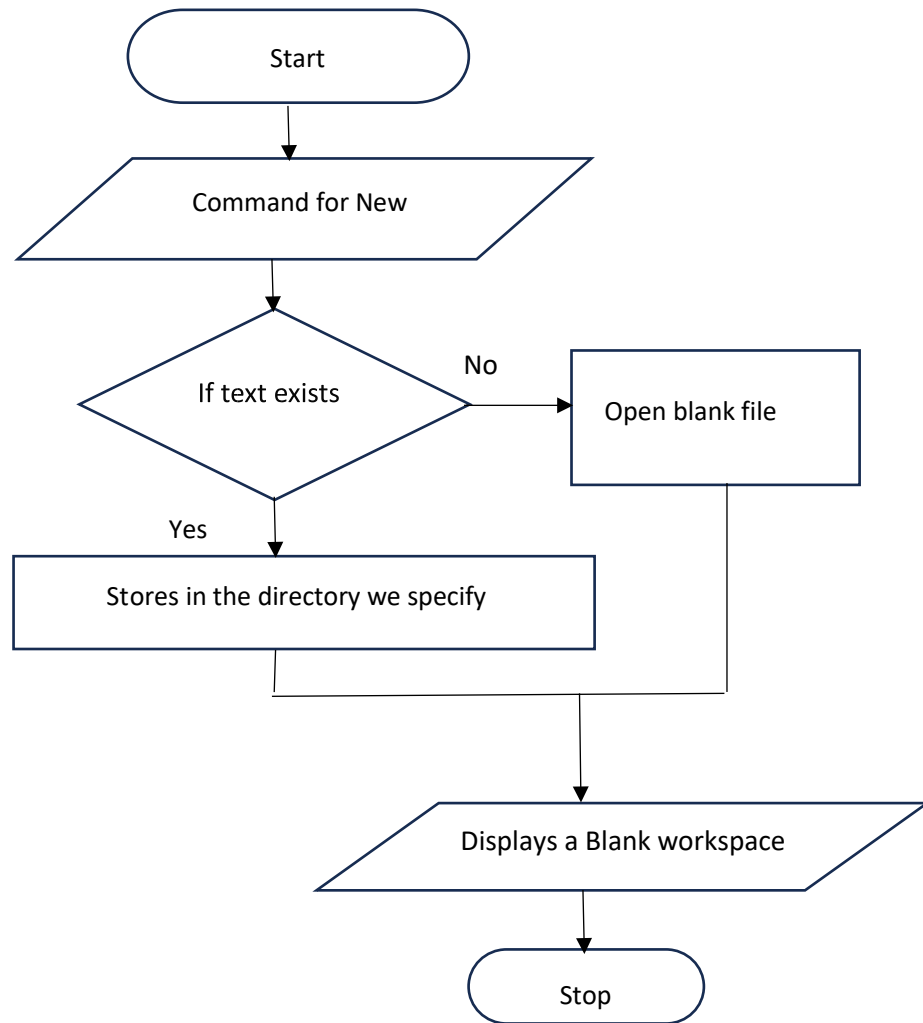
Flowchart for Redo:



Flowchart for Undo:



Flowchart for New:



Algorithm For NEW:

Step1: Start

Step2: Choose new option.

Step3: If file exists

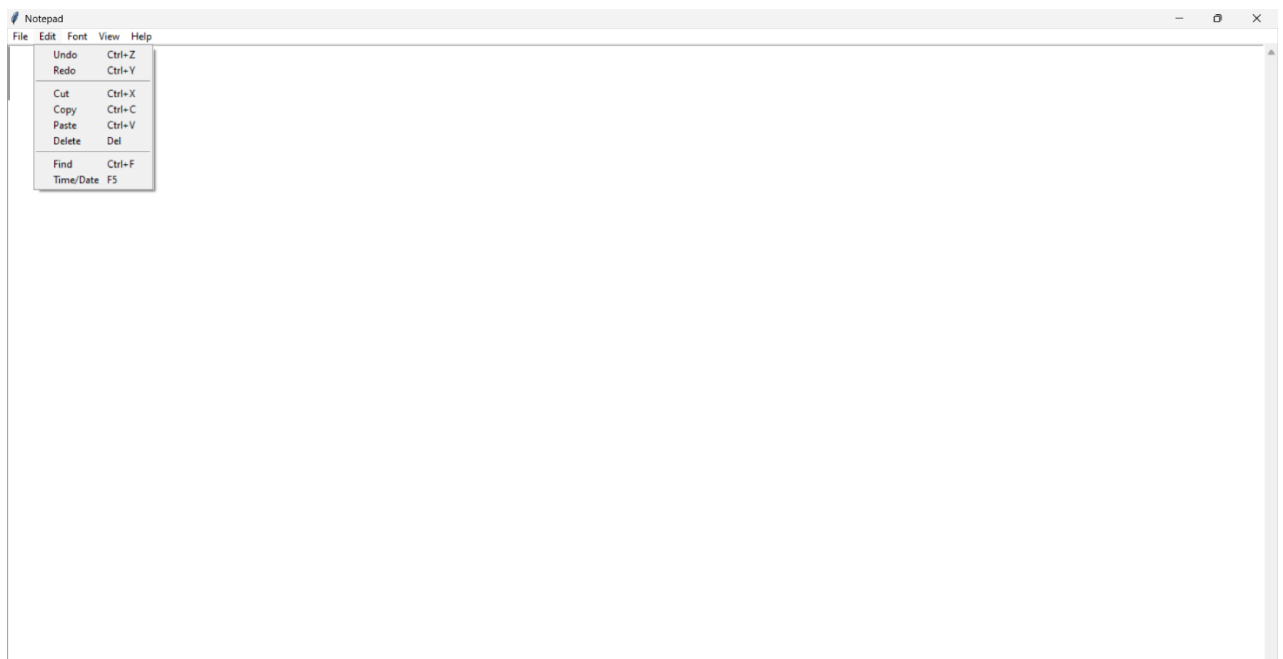
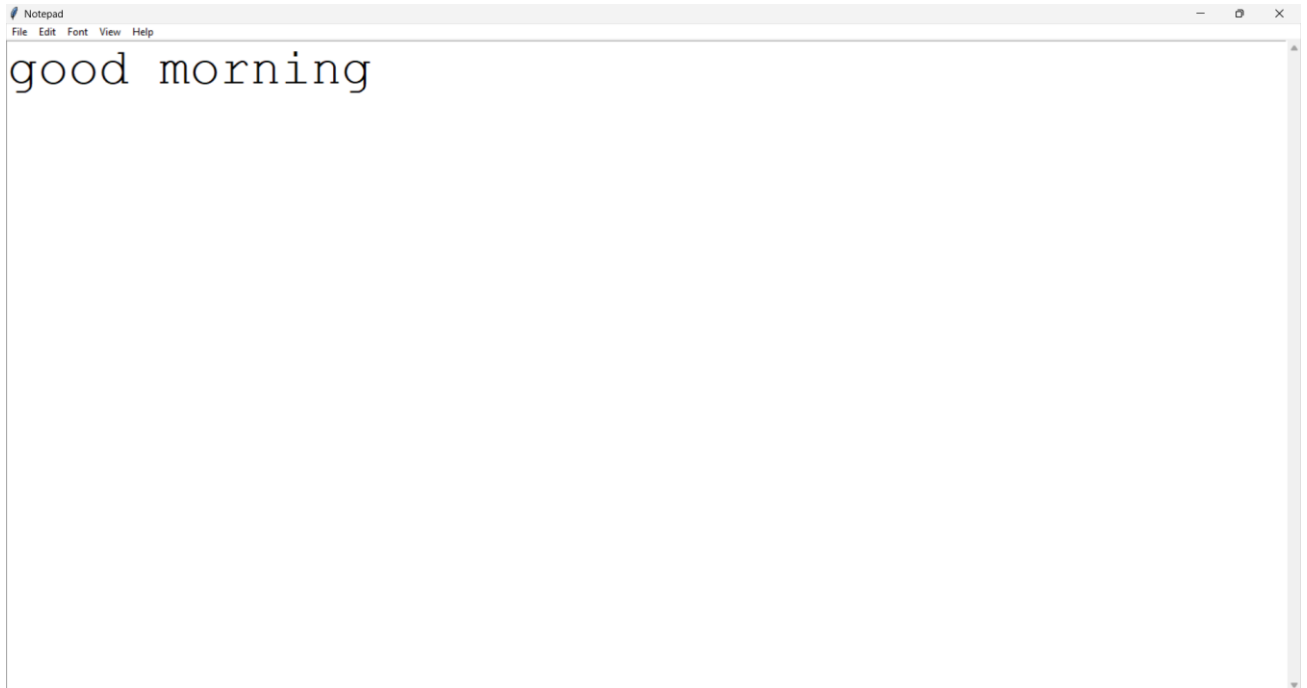
Stores in internal storage

End If

Step4: Display blank file.

Step5: Stop

4.User Interface



5.TESTING

5.1 Introduction

Testing is the major quality control measures and during the software development it is used to detect errors that could have occurred during any of the phase like requirement analysis, design, coding. The goal of the testing is to uncover errors in the program.

5.2 Levels of Testing

Testing is done in different levels which includes the following.

- Unit Testing
- Integration Testing
- System testing
- Acceptance testing

- **Unit Testing**

In Unit testing each module gets tested during the coding phase itself. The purpose is to exercise the different parts of the module code to detect the coding errors.

- **Integration Testing**

After new testing the modules are gradually integrated into sub systems. It is performed to detect design errors by focusing on testing the interconnection between modules.

- **System Testing**

System is tested against the system requirement if all the requirements are met and if the system performs as specified by the requirement.

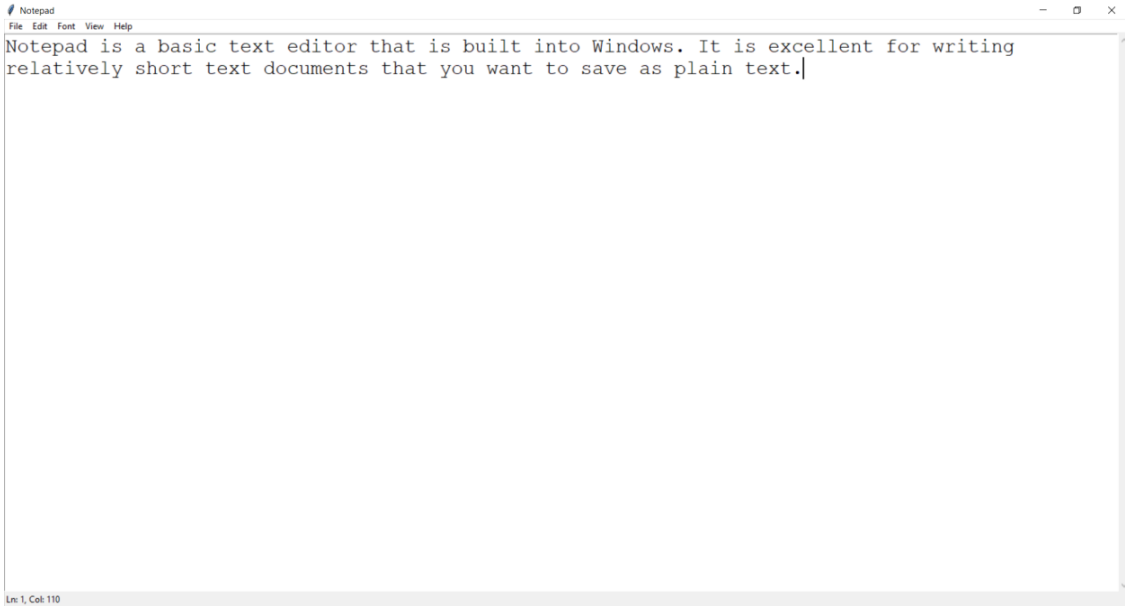
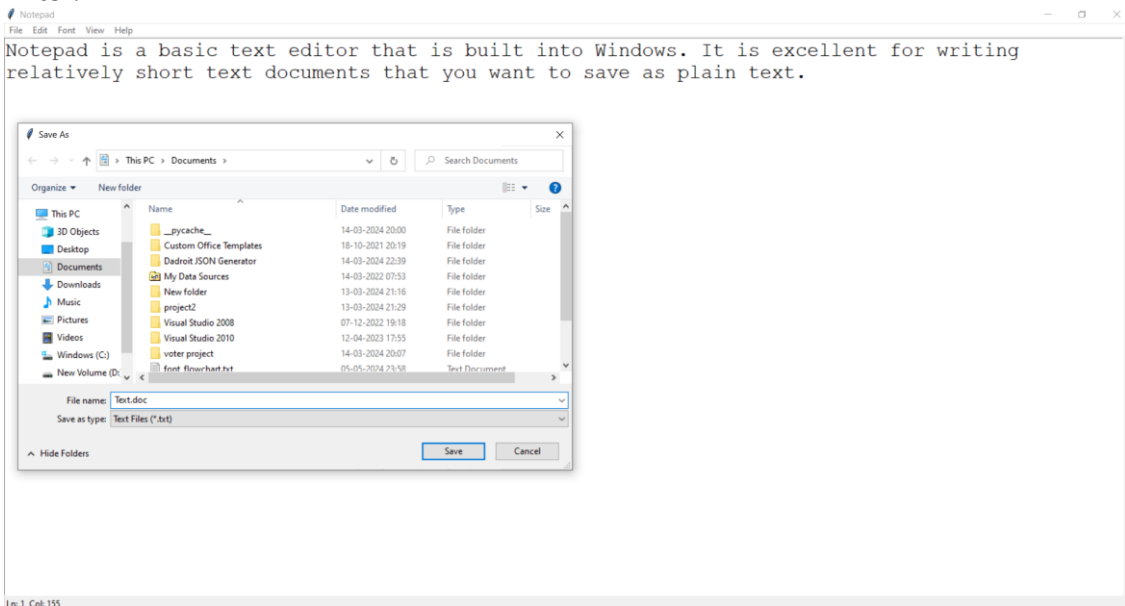
- **Acceptance Testing**

It is performed to demonstrate to the client on real life data of the client, the operation of the system.

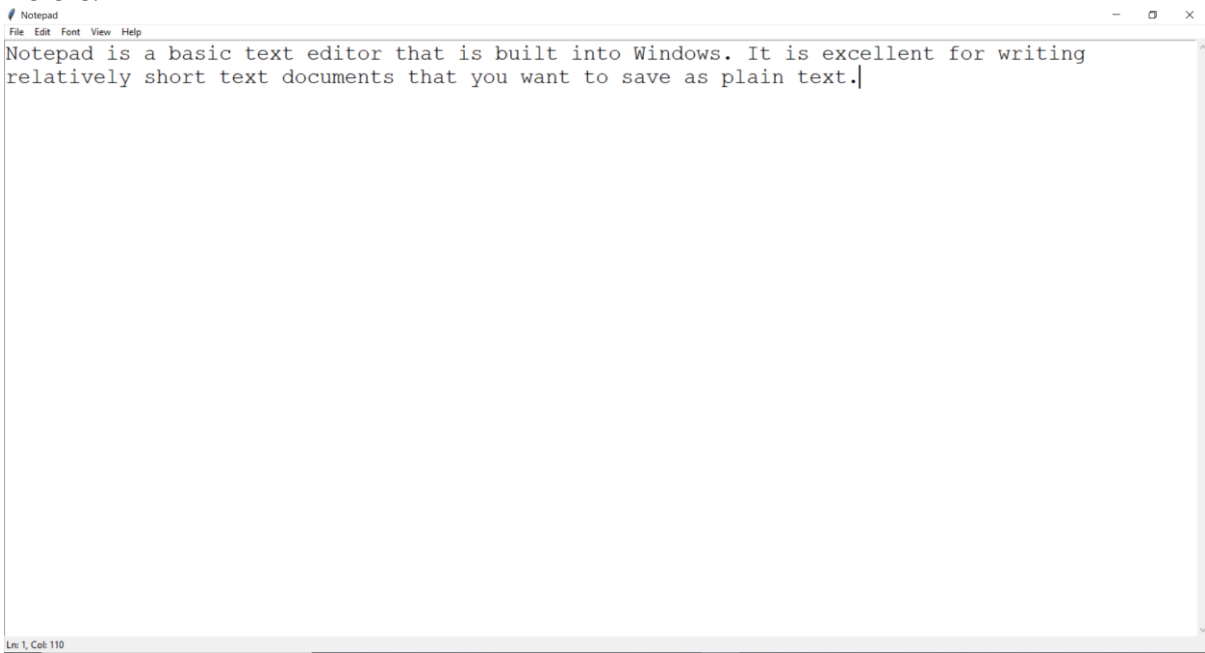
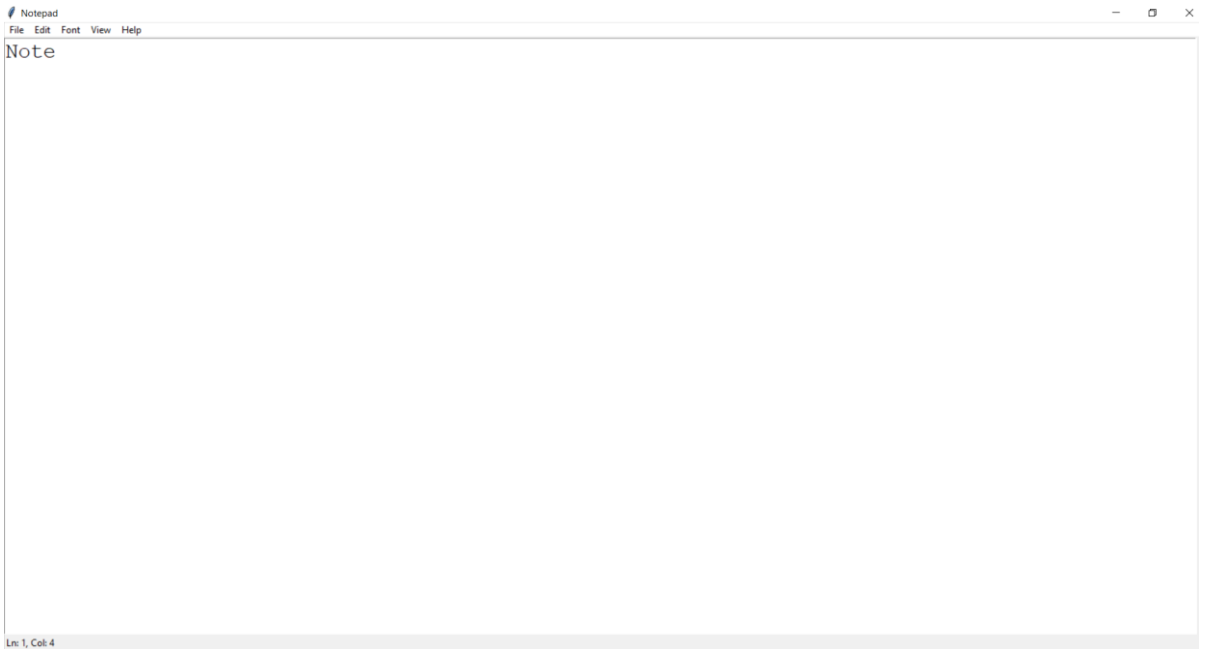
5.3 Test Case

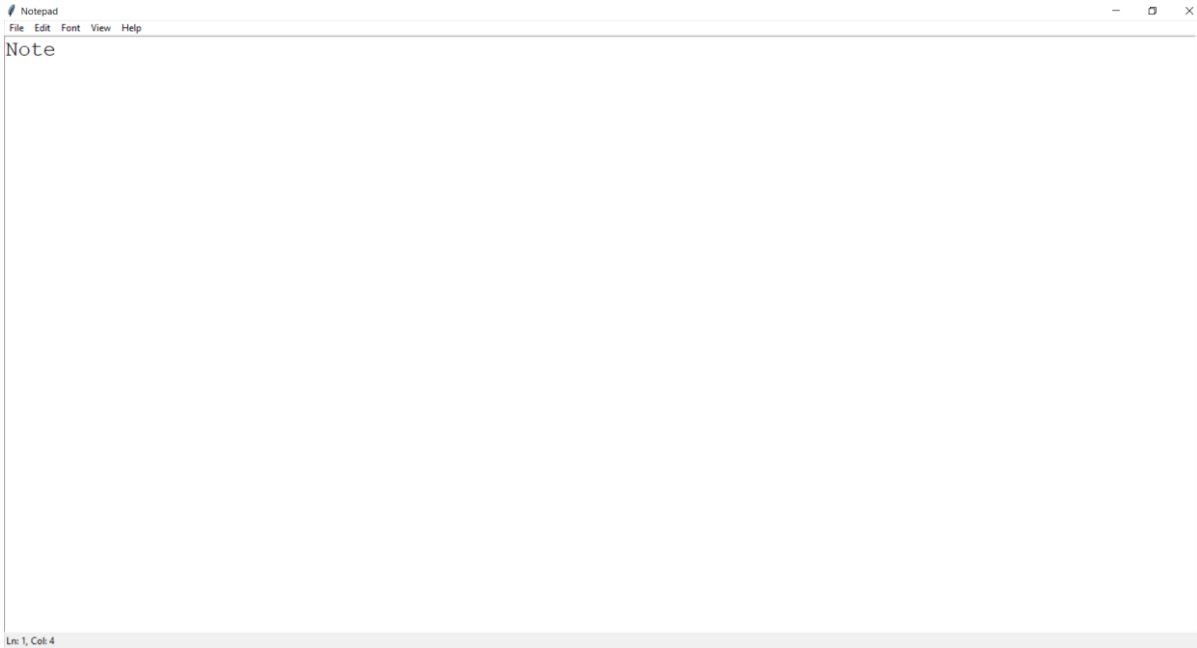
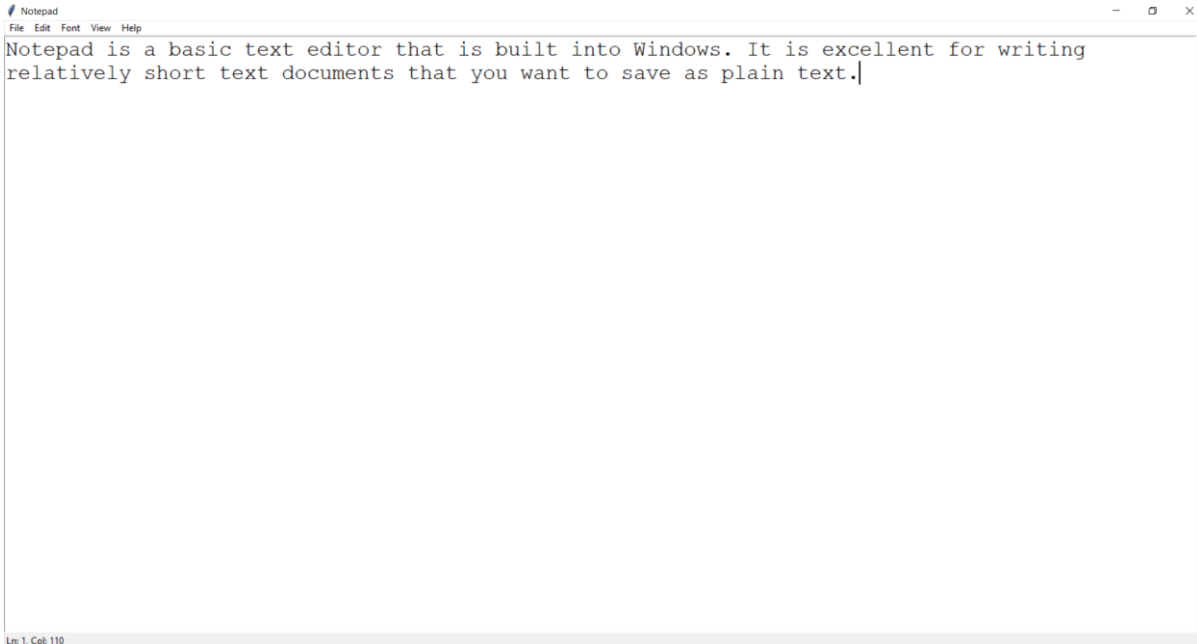
It is the input that tests the genuineness of the program and successful execution Of the test case reveals. that there are no errors in the program that are under testing. It is a set of conditions or variables under which tester will determine whether an application or software is working currently

Test case ID	01
Title	Save
Purpose	To Save the file

Test data	Text (Input)
Steps	1. Open the notepad application. 2. Type the required text in the text area provided. 3. Save the text typed from the save option of the file module.
Expected output	Before:  After: 

Test case ID	02
Title	Undo
Purpose	Undo the last action
Test data	Text

Steps	1. Click on the undo option. 2. Undo the last typed texts. 3. Display the texts prior to the current action.
Expected output	Before:  The screenshot shows the Notepad application window. The title bar reads 'Notepad'. The menu bar includes 'File', 'Edit', 'Font', 'View', and 'Help'. The text area contains the following text: 'Notepad is a basic text editor that is built into Windows. It is excellent for writing relatively short text documents that you want to save as plain text.' The status bar at the bottom indicates 'Ln: 1, Col: 110'. After:  The screenshot shows the Notepad application window after the undo action. The title bar reads 'Notepad'. The menu bar includes 'File', 'Edit', 'Font', 'View', and 'Help'. The text area now only contains the word 'Note'. The status bar at the bottom indicates 'Ln: 1, Col: 4'.

Test case ID	03
Title	Redo
Purpose	To recover the undone action.
Test data	Texts
Steps	1. Click on the redo option. 2. Redo the last action. 3. Recovering undone action.
Expected output	Before:  After: 

6.BIBLIOGRAPHY

- Python Crash Course by Eric Matthes.
- Python project for Beginners by Connor P.Milliken.
- Automate the Boring Stuff with Python by Al Sweigarl.

